

Incident Response, Operations, and Disaster Recovery

A comprehensive guide to detecting threats, responding to incidents, and ensuring rapid recovery from disruptions



Course Overview

What You'll Master

This course equips you with essential skills to detect security threats, respond effectively to incidents, implement robust backup strategies, and ensure your organisation recovers swiftly from any disruption.

01

Incident Response Management

Master the complete lifecycle from detection to recovery

02

Logging and Monitoring

Implement comprehensive audit trails and detection systems

03

Data Recovery and Business Continuity

Ensure rapid restoration and operational resilience

Who This Is For

Students and entry-level IT and security professionals seeking hands-on incident response and disaster recovery expertise to launch their cybersecurity careers.

What Is Incident Response?

Incident Response is your organisation's strategic game plan for handling security breaches, cyberattacks, and system failures. Think of it as your emergency response protocol—similar to a fire drill, but designed specifically for cybersecurity threats.

 **Critical Insight:** Just as fire drills save lives through preparation, a well-rehearsed incident response plan can save your organisation from catastrophic losses. The difference between a contained incident and a devastating breach often comes down to preparation.



Building Your Incident Response Plan



Preparation

Establish tools, train teams, and create procedures before incidents occur



Detection

Identify when security events or system failures happen



Analysis

Understand the scope, impact, and severity of the incident



Containment

Stop the problem from spreading to other systems



Recovery

Restore systems and services to normal operations



Post-Incident Review

Document lessons learnt and improve future responses

Real-World Scenario: Website Down

Imagine your company's website suddenly becomes inaccessible. Your incident response plan must immediately answer these critical questions:

Who gets notified first?

Define clear escalation paths and contact protocols for immediate notification

What logs should you check?

Identify which system logs, access records, and traffic patterns to examine

Attack or technical glitch?

Establish criteria to distinguish between malicious activity and system failures

How do you isolate affected systems?

Document specific containment procedures to prevent spread

How do you communicate with customers?

Prepare messaging templates and communication channels

Assigning Response Team Roles



Incident Response Lead

The primary decision-maker coordinating the entire response



Backup Lead

Steps in when the primary lead is unavailable



Technical Analysts

Investigate incidents and implement technical fixes



Communications Coordinator

Manages internal and external messaging



Legal/Compliance Officer

Ensures regulatory requirements are met

❏ **Key Point:** Even if you engage an external security provider, you must maintain at least one internal person who oversees their work and understands your specific business priorities.

Training and Testing Your Team

Regular training exercises are essential to ensure your team responds effectively under pressure. Different exercise types serve different purposes:

Tabletop Exercises

Sit-down discussions walking through hypothetical scenarios to test decision-making processes

- Focus on procedures and communication
- Low-cost, easy to conduct
- Ideal for identifying gaps in plans

Simulations

Technical drills that test actual systems, tools, and procedures in controlled environments

- Validate technical capabilities
- Moderate complexity and cost
- Reveals tool and access issues

Full-Scale Tests

Realistic scenarios involving all team members responding as they would in actual incidents

- Most comprehensive validation
- Higher cost and coordination needs
- Builds team confidence and muscle memory

Recommended Frequency: Conduct exercises at minimum annually. High-risk organisations should train quarterly to maintain readiness.

Sample Tabletop Exercise

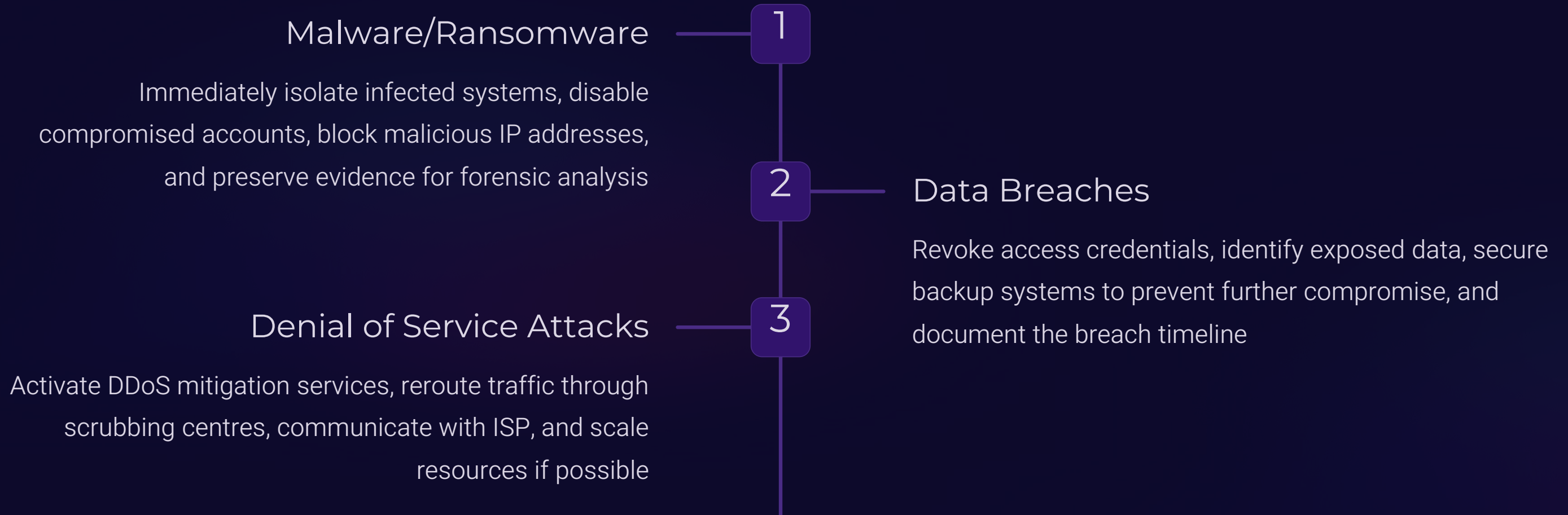
"It's 2 AM on Saturday. Your monitoring system alerts you that someone is attempting to access your customer database from an IP address in Russia. The attempts started 30 minutes ago."

What This Exercise Reveals

- Can team members access the IR plan after hours?
- Do they have the right tools and permissions?
- Are contact lists current and accessible?
- How quickly can critical decisions be made?
- Are communication channels established?
- Do team members know their specific roles?



Containment Strategies by Incident Type



📄 Every containment action should be documented with precise timestamps. This evidence trail is critical for post-incident analysis and potential legal proceedings.

Post-Incident Reviews: Learning from Every Event



Within 1–2 weeks after an incident, conduct a structured review meeting. This is where real improvement happens.

1 Timeline Reconstruction

Create a detailed chronology of what happened and when

2 Response Evaluation

Assess what worked well and what didn't during the response

3 Root Cause Analysis

Identify why the incident occurred in the first place

4 Impact Assessment

Calculate financial, reputational, and operational costs

5 Improvement Actions

Define specific changes to prevent recurrence



Logging and Monitoring

Logging records what happens on your systems—who logged in, what files were accessed, which commands were executed. Monitoring actively watches those logs to detect problems in real time.

The Analogy: Logs are your security cameras; monitoring is having someone watch those cameras 24/7.

Creating Your Audit Log Management Process

1

Log Collection Requirements

- Which systems must generate logs
- What information each log must contain
- How logs are collected and stored
- Who has access to logs

2

Log Review Requirements

- Who reviews logs and how often
- What to look for (failed logins, privilege escalations)
- How to escalate suspicious findings

3

Log Retention Requirements

- How long to keep logs (minimum 90 days)
- Where logs are stored (centralised, encrypted, backed up)
- Payment systems require 12 months

4

Time Synchronisation

- All systems use the same time source (NTP)
- Critical for correlating events across systems
- Enables accurate timeline reconstruction

Why Time Synchronisation Matters

Without Synchronised Time

Events appear disconnected and the attack sequence is unclear:

- Server A: Login at 2:15 PM
- Server B: File deletion at 2:10 PM
- Firewall: Suspicious connection at 2:20 PM

You cannot determine the order of events!

With Proper Synchronisation

A clear attack timeline emerges:

- 2:15:03 PM – Suspicious login detected
- 2:15:47 PM – User escalated to admin privileges
- 2:16:22 PM – Large file downloaded
- 2:16:55 PM – Connection to suspicious external IP

Clear evidence trail for investigation!

 **Critical Statistic:** In 90% of data breaches, attack evidence existed in logs but wasn't noticed. Proper log management can detect breaches in days instead of the average 207 days.

Essential Log Fields and Data Types



Timestamp

Exact date and time down to the second



User Identity

Username, employee ID, or account details



Source IP Address

Where the connection originated from



Action Performed

What happened (login, file access, command)



Result

Success or failure of the action



Data Affected

What files, records, or resources were involved

Special Log Types to Monitor

DNS Query Logs: Track websites accessed to detect malware calling home to command-and-control servers

Command-Line Audit Logs: Record commands executed on servers to detect privilege abuse

Application Logs: Track login attempts, data exports, and configuration changes

Centralising and Analysing Logs

Manual log review is impossible at scale. A medium-sized company generates millions of log entries daily. Centralisation with automated analysis helps you find the one malicious action amongst millions of legitimate ones.

Centralisation Solutions



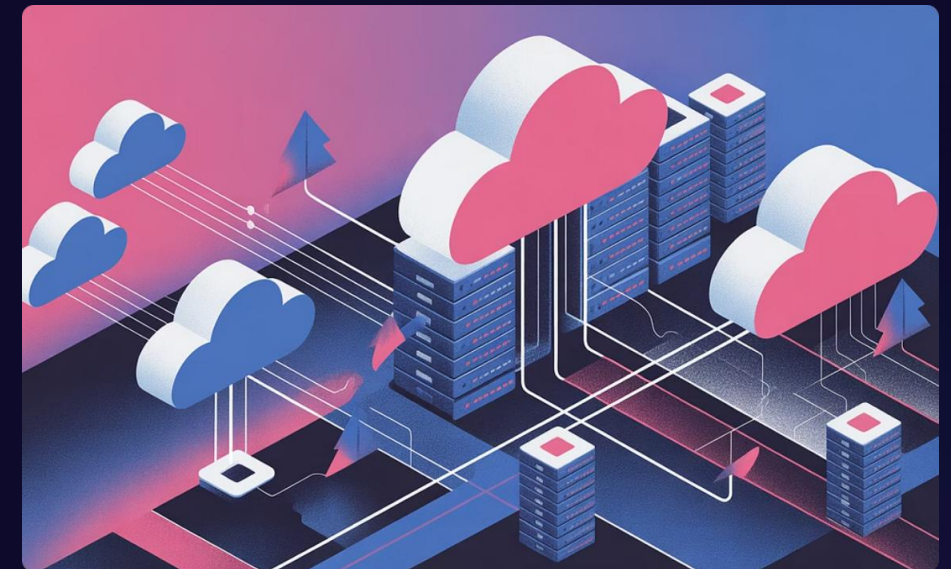
SIEM Tools

Commercial solutions like Splunk, IBM QRadar, or Microsoft Sentinel



Open-Source Platforms

ELK Stack (Elasticsearch, Logstash, Kibana) and Graylog



Cloud Logging Services

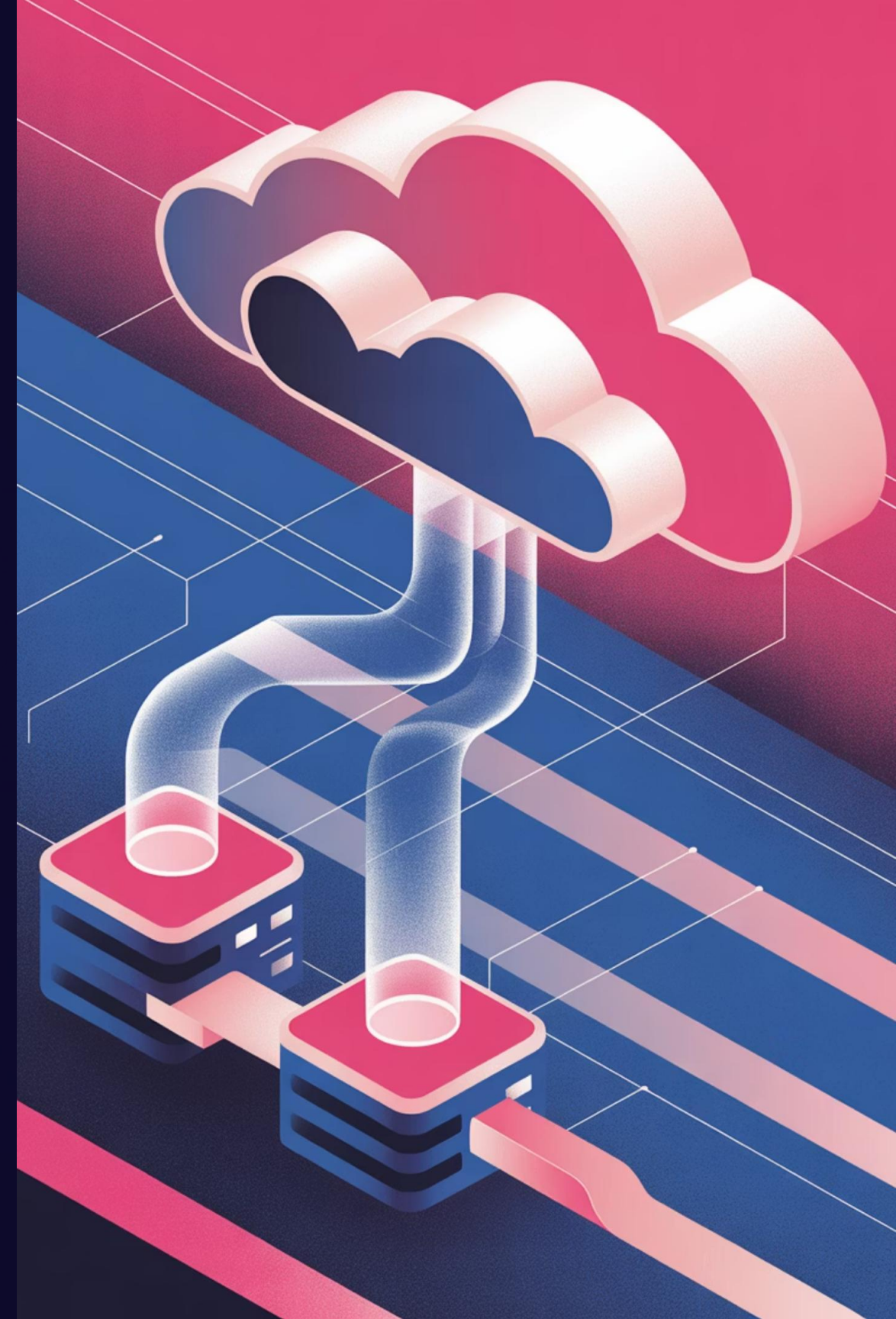
AWS CloudWatch, Azure Monitor, Google Cloud Logging

Automated Alert Examples

- 10+ failed login attempts from same IP within 5 minutes (brute force attack)
- Any login from a country where you have no employees or offices
- Database queries returning more than 10,000 records (potential data theft)
- Admin account used outside business hours

Data Recovery and Business Continuity

Business Continuity ensures your organisation continues operating even when disaster strikes—whether that's a cyberattack, natural disaster, hardware failure, or human error.



Planning Your Data Recovery Process

Not all data is equally important. Prioritisation ensures you recover critical systems first, minimising financial impact.



Tier 1: Critical

RTO: 4 hours | RPO: 15 minutes

Payment processing, customer-facing services. Every hour of downtime costs thousands in lost revenue.



Tier 2: Important

RTO: 24 hours | RPO: 4 hours

Email, internal applications. Impacts operations but not immediate revenue.



Tier 3: Standard

RTO: 72 hours | RPO: 24 hours

Archived files, development systems. Important but not time-critical.

RTO (Recovery Time Objective): How quickly must each system be restored?

RPO (Recovery Point Objective): How much data loss is acceptable?

Implementing Automated Backups

Full Backup

Complete copy of all data

Pros: Simple to restore, complete protection

When: Weekly or monthly for complete systems

Incremental Backup

Only changes since last backup

Pros: Fast, efficient storage

When: Daily, combined with weekly full backups

Differential Backup

Changes since last full backup

Pros: Faster restoration than incremental

When: Alternative for medium-sized datasets

The 3-2-1 Backup Rule



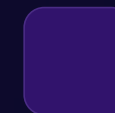
3 Copies of Your Data

Original plus two backups



2 Different Media Types

Local disk plus cloud storage

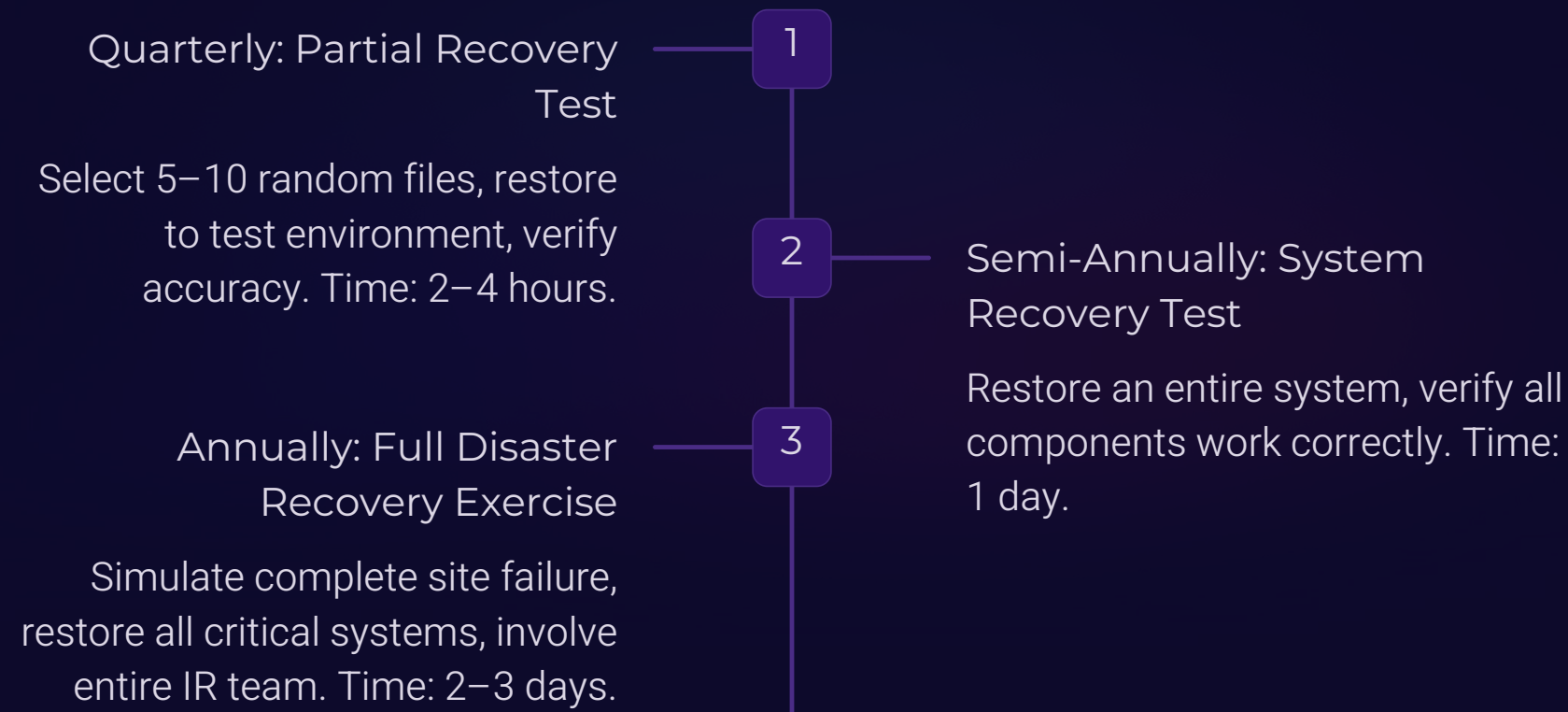


1 Copy Off-Site

Protects against physical disasters

Testing and Verifying Recovery

Untested backups are just files on a disk. Studies show 34% of companies never test their backups, and 77% of those suffer complete data loss when they try to restore.



❏ **Critical Success Factor:** Companies with tested recovery plans restore operations 3x faster than those without. Testing identifies problems when the stakes are low.



Your Path Forward

Preparation

You cannot respond effectively without advance planning. Document everything and test your plans regularly.

Improvement

Every incident is a learning opportunity. Update plans based on lessons learnt.



Detection

Comprehensive logging is your early warning system. Automated monitoring catches threats humans would miss.

Response

Speed matters—every minute counts. Follow your plan but adapt to circumstances.

Recovery

Backups are your insurance policy. Test them regularly and prioritise critical systems first.

Remember: Incidents are inevitable—preparation is optional. The best time to prepare for an incident is now, before one happens.